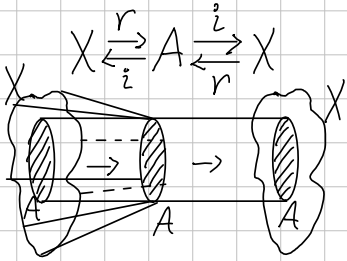
Def. Let X be a space, and $A \subset X$ be a subspace with inclusion $i: A \hookrightarrow X$.
A continuous map $r: X \to A (\subset X)$ is called a retraction if $r \circ i = id_A$

In the Algebraic Topology, we say that i for $A \subset X$,

X deformation retract if there is a retraction $r: X \to A$ such that $i \circ r \simeq id_X$



Thm. Let $A \subset X$, $i : A \hookrightarrow X$ be an inclusion, and $f : X \to A$ be a continuous. Suppose that $i \circ f \simeq id_X$ via the Homotopy $H : X \times I \to X$.

[Prop. The Möbius band M is Homotopy equivalent to the cylinder $S^1 \times I$.